



$$g(z) = \frac{1}{1+e^{-z}} \quad \text{und} \quad g'(z) = g(z)(1-g(z)) \quad : \text{Bsp 1.1}$$

$$g'(z) = \left(\frac{1}{1+e^{-z}} \right)' = \frac{e^{-z}}{(1+e^{-z})^2} = \frac{1}{1+e^{-z}} \cdot \frac{e^{-z}}{1+e^{-z}} = g(z) \cdot \left[1 - \frac{e^{-z}}{1+e^{-z}} \right] =$$

$$= g(z) \cdot (1-g(z))$$

$$\frac{\partial \ell(\omega)}{\partial \omega} = \frac{1}{N} \sum_{t=1}^N (y_t - g(\omega x_t)) \vec{x}_t \quad : \text{Bsp 1.2}$$

: Bsp 1.3

$$\ell(\omega) = \frac{1}{N} \sum_{t=1}^N \log p(y_t | x_t)$$

$$\frac{\partial \ell(\omega)}{\partial \omega} = \frac{1}{N} \sum_{t=1}^N \frac{\partial \log p(y_t | x_t)}{\partial \omega} = \frac{1}{N} \sum_{t=1}^N \frac{1}{p(y_t | x_t)} \cdot \frac{\partial p(y_t | x_t)}{\partial \omega}$$

: Bsp 1.4

$$p(y | x) = \begin{cases} p(y=1 | x) = g(\omega x) = g(\omega x)^1 \cdot [1-g(\omega x)]^{1-1} = g(\omega x)^1 \cdot [1-g(\omega x)]^{1-1} \\ p(y=0 | x) = 1-g(\omega x) = g(\omega x)^0 \cdot [1-g(\omega x)]^{1-0} \end{cases}$$

$$\frac{\partial p(y | x)}{\partial \omega} = \frac{\partial [g(\omega x)^y \cdot [1-g(\omega x)]^{1-y}]}{\partial \omega} = y \cdot g(\omega x)^{y-1} \cdot \frac{\partial g(\omega x)}{\partial \omega} + g(\omega x)^y \cdot (1-y) [1-g(\omega x)]^{-y} \cdot - \frac{\partial g(\omega x)}{\partial \omega} =$$

$$\frac{\partial g(\omega x)}{\partial \omega} = \frac{\partial g(\omega x)}{\partial (\omega x)} \cdot \frac{\partial (\omega x)}{\partial \omega} = g(\omega x) [1-g(\omega x)] \cdot x$$

$$\begin{aligned} \frac{\partial p(y | x)}{\partial \omega} &= y \cdot g(\omega x)^{y-1} \cdot [1-g(\omega x)] \cdot x - g(\omega x)^y \cdot (1-y) \cdot [1-g(\omega x)]^{-y} \cdot x = \\ &= g(\omega x)^{y-1} \cdot [1-g(\omega x)] \cdot x \cdot [y \cdot [1-g(\omega x)]^y - g(\omega x) (1-y)] \end{aligned}$$

$$\frac{1}{p(y|x)} \cdot \frac{\partial p(y|x)}{\partial \omega} = x \left[y \cdot [1 - g(\omega x)]^y - g(\omega x) (1 - y) \right] =$$

$$= \begin{cases} y=1 \rightarrow x[1 - g(\omega x)] = x[y - g(\omega x)] \\ y=0 \rightarrow x[-g(\omega x)] = x[y - g(\omega x)] \end{cases} = x[y - g(\omega x)]$$

$$; \frac{\partial \ell(\omega)}{\partial \omega}$$

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$$\frac{\partial \ell(\omega)}{\partial \omega} = \frac{1}{N} \sum_{t=1}^N \frac{1}{p(y_t|x_t)} \cdot \frac{\partial p(y_t|x_t)}{\partial \omega} = \frac{1}{N} \sum_{t=1}^N \vec{x}_t [y_t - g(\omega x_t)]$$